

Lesson 6.14: Build the Master Flowchart

Unit: Unit 6, From Farm to Table (Food Systems and Production) **Topic:** 6.3 Feed the Class project **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes **Room:** FACS lab with kitchen stations (teams work at the station they will cook in; no cooking today)

Standards

- NYS module line: NYS FSP 5. Consumer Resources and Finance d) Apply time management, organizational, and process skills to prioritize tasks and achieve goals
- NYS module line: NYS FSP 1. Basic Culinary Skills a) Apply basic culinary and management skills to the preparation of nutritious food
- NYS module line: NYS FSP 1. Basic Culinary Skills c) Demonstrate safety and sanitation procedures when handling food and tools (written into the station card today; demonstrated in Lesson 6.16)
- CTE theme line: CTE CIR 3. Workplace Communication f) Demonstrate effective communication skills in a group setting to accomplish a task
- CTE theme line: CTE CIR 3. Workplace Communication e) Explain how to give and receive accurate reports in a variety of formats
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students are able to budget their time and money.
- CDOS Standard 3a Universal Foundation Skills: Managing Resources. Students understand the material, human, and financial resources needed to accomplish tasks and activities.
- CDOS Standard 3a Universal Foundation Skills: Managing Information. Students select and communicate information in an appropriate format (e.g., oral, written, graphic, pictorial, multimedia).
- National FCS 3.0: 2.1.1 Use time management, organization, and step-by-step process skills to rank tasks and reach goals.
- National FCS 3.0: 8.4.3 Work out the food, equipment, and supplies a menu needs. (Confirmed in the 3.0 text in 02 Standards/STANDARDS SOURCES - web research.md, Area 8; not printed in the FSP block, so it is a secondary citation.)

Enduring understanding and essential question

Big idea: In a kitchen, time is the ingredient you cannot buy more of. A plan that says who does what in which five minutes is the difference between a meal and a mess. And one step always holds up everything else. **Essential question:** What is the one step everything in our station waits on, and how do we build the clock around it?

Objectives

By the end of this lesson, students will be able to:

1. (Low) Given their recipe and a blank eight-block flowchart, place every step of the recipe into a five-minute block from setup through clean up, with a person's role named on each step.
2. (Mid) Identify the critical path in their own plan: the single step everything else waits on, and explain what happens to the whole period if that step starts five minutes late.
3. (High) Write a station card in the room's standard format (ingredients for the station, ten numbered steps, safety reminders, and a "what does done look like" checklist) that another team could follow without asking a question.

Vocabulary

Tier 2 (general academic): sequence, simultaneous, prioritize, bottleneck Tier 3 (FACS): flowchart, critical path, station card, mise en place, prep, hold time. See the unit vocabulary list for the Spanish column.

Materials and setup

- Handout, one per student plus one poster-size blank per station: Handout 06.14 - Flowchart Builder and Station Card Template.md (the eight-block flowchart grid, the critical path finder, the ten-step station card template, the safety reminder bank, and the "what does done look like" checklist)
- Each station's signed costing sheet and scaling worksheet from Lesson 6.13
- One poster-size sheet per station (11 by 17 or larger) and a marker per station
- Sticky notes, a pad per station, one step per note
- A timer, and the room clock visible
- The class master flowchart wall spot: four columns taped out on the whiteboard or on butcher paper, one per station, with the eight time blocks labeled down the left
- Slides: Slides 06.14 - Build the Master Flowchart (outline in 05 Materials; the site renders it as a deck)
- Setup notes: tape out the four-column master grid before the bell and label the blocks 0 to 5, 5 to 10, 10 to 15, 15 to 20, 20 to 25, 25 to 30, 30 to 35, 35 to 40. Put each station's Lesson 6.13 papers on its own counter. Post the allergy list again. No food today.

Pre-assessment

Every station already ran a three-lab cycle in Topic 6.2 with a flowchart handed to them (Lessons 6.7, 6.8, and 6.9). The do now asks them to remember what went wrong in one of those labs, which tells you which stations already know that time is the problem and which think the recipe is the problem.

Do now and hook (Time: 5 min)

On the board: “Think about the pasta lab or the quesadilla lab. Write the one moment your station was standing around waiting. What were you waiting for?”

Debrief: take four answers, one per station, and write them in a column. In almost every class the list is the same three things: waiting for water to boil, waiting for the oven, waiting for the teacher. Circle those three. Then the hook: “Those are not accidents. Those are the steps everything else waits on, and a real kitchen builds the clock around them. Today you build the clock.”

Say the line that frames the period: “I ran four kitchens in one period with a flowchart on one page, in five-minute blocks, and it was written so a substitute could run it. That is what you are making.”

Procedure

Step	Teacher will	Students will	Time
1. Direct instruction: what a five-minute block flowchart is	Show a completed example on the slide: one column, eight blocks, two or three short lines per block, a role name on each line, from setup at 0 to 5 through clean up at 35 to 40. Name the five phases out loud: setup, prep, cook, serve and eat, clean. Say the rule: every block has something for every person; nobody has a blank block. Point out that the block labeled 20 to 25 on the example says KEY CHECK, because that is the moment somebody finds out whether the dish is on time.	Copy the eight block labels onto the flowchart builder. Write the five phase names.	5 min
2. Learning activity: sticky notes into blocks	Teams break their recipe into steps, one step per sticky	Write one step per sticky note. Place every note into a	12 min

	note, then place the notes into the eight blocks on the poster. Rule: read the step, ask “how long does this take,” then ask “who does it.” Circulate and ask one question per station: “What is happening in your 15 to 20 block? Show me.”	block. Write a role name on every note. Move notes until every block has work for every person.	
3. Find the critical path	Teach the critical path in one move: “Point to the step that, if it is late, makes everything after it late. That is your critical path step.” Model it on the pasta station: the water boiling. Nothing else on that station can happen until it boils, so the water goes on in block one, before anyone washes a measuring cup. Then give the test question: “If that step starts five minutes late, what time do you eat?” Every station writes the answer.	Circle the critical path step on the poster in a second color. Fill the critical path finder: the step, why everything waits on it, the earliest block it can start, and what happens to the eat time if it slips five minutes.	7 min
4. Write the station card	Hand out the station card template and read the four parts: ingredients for this station, ten numbered steps with one action each, safety	Write the station card, ten numbered steps, from the flowchart. Add three safety reminders from the bank plus one that is specific to this dish. Fill the	9 min

	reminders, and “what does done look like.” Give the test that decides whether the card is finished: another team has to be able to run it with no questions. Pair the stations and have each read the other’s card and write one question it could not answer.	four “done looks like” boxes. Swap with the paired station and write one question.	
5. Post to the master grid	Call each station up to copy its eight blocks into its column on the class master grid, shortest phrases only. Then read the grid across, block by block, out loud, and stop at the collisions: two stations on the oven at once, two stations needing the teacher to drain at the same minute, everybody plating in the same two minutes. Fix the collisions with the class: one station shifts a block. Write the fixes in red.	One person per station posts. Everyone watches the read-across and writes the fix on their own copy.	2 min

Do now 5 + steps 35 = 40, with the 2-minute closure inside Step 5. This is the tightest lesson in the topic. If it runs short, cut Step 4’s paired read to 5 minutes and let the station card finish as the do now of Lesson 6.15; never cut the read-across in Step 5, because that is the part that saves cook day.

Questions to ask

Monitor understanding:

- What is happening in your 10 to 15 block? Name the person doing it.
- Which of your steps takes the longest? Is it in an early block or a late one?
- Your Prep Cook has nothing in blocks four and five. What are they doing?
- If the oven is not preheated until minute 12, what time does your dish come out?
- Read me step 6 of your card. Is that one action or three?

Deepen learning:

- A restaurant calls getting everything measured and laid out before you start mise en place. Why would a cook do that instead of measuring as they go?
- Two stations need the oven. In a real kitchen who decides who gets it, and what do they decide with?
- Your critical path is the water boiling. Name one thing you could do to shorten it. (Start with hot tap water. Cover the pot. Use less water.)
- Your card says “cook until done.” Would a substitute teacher know what that means? What number or picture would fix it?
- What is the one thing on your flowchart that only I can do, and what happens to your station if I am at another station when you need me?

Check for understanding

During Step 2 and Step 3, a clipboard walk with a yes or no per station on three items: every block has a role name in it, the critical path step is circled and is actually the bottleneck, and the “if it slips five minutes” answer is a real time and not “we would be late.” During Step 4, the paired station’s one question is the check on the station card: a card that draws a question about a step is not done, and the team fixes that step before the bell.

Closure (Time: 2 min, inside Step 5)

Each station reads one line aloud: “Our critical path is , **it starts at minute** , and if it slips we eat at ____.” Then the whole class reads the 30 to 35 block across all four columns out loud, because that is the serve block and it is the one that has to be right. The poster stays on the wall through Lesson 6.17. The station card goes in the project folder and a clean copy gets laminated or sleeved for cook day.

Differentiation and supports

- ELL: the flowchart is a picture document by design, which is why it works; each block on the poster gets a drawn symbol (a flame, a clock, a plate, a sponge) next to the words; the station card template has a picture flowchart version on page 4, eight boxes with room to draw the step and one word under it, and a team may submit the picture version as its real card; the safety reminder bank is printed in Spanish as well as English; Vocabulary 06.md carries flowchart, critical path, station card, and prep in English and Spanish.
- IEP and 504 (general): sticky notes are the accommodation, not an extra, because a student who cannot draft in order can write steps in any order and sort them; the ten-

step template accepts eight steps if each one is complete; a student may dictate a step to a teammate who writes it; the poster work is standing or seated; the paired read in Step 4 gives a student who reads slowly a partner and a purpose.

- Grade 6 support: the flowchart grid comes with the setup block and the clean block already filled in from the room's standard routine, so the team plans only the middle six blocks. The station card may be eight steps.
- Grade 8 stretch: the team writes a second version of the flowchart for a 30-minute period and says which two steps get cut and what the dish loses. A second stretch: the team writes the substitute-teacher version of its card, which means every step has a number, a time, and a "what to do if this goes wrong" line.
- UDL checkpoint used: multiple means of action and expression (a plan can be built as sticky notes, a grid, or a drawn picture chart, and all three count) and multiple means of representation (the same schedule appears as a team poster, a class master grid, and a written card).
- No-kitchen alternative: the lesson is planning only, so it runs anywhere. For a no-kitchen room, the flowchart blocks change from cook phases to assembly phases (setup, portion, assemble, serve and eat, clean) and the critical path becomes whatever has to be opened, drained, or brought to the table first. The station card template is the same.

Allergy line

No food is handled. The station card has a required allergen box, and a card is not approved without it. Every station copies the allergens in its own dish from its costing sheet onto the card, and the Station Manager reads the box aloud at Step 5. This is the card that gets read to the class before anybody eats in Lesson 6.17.

Assessment

Objective	Standards met	Learning task	Assessment (formative or summative; tool)
1. Place every step into a five-minute block with a role named	FSP 5. d); CIR 3. f); Std 3 PI C; National FCS 2.1.1	Sticky notes into blocks, Step 2	Formative; clipboard yes or no on "every block has a role name"
2. Identify the critical path and predict the slip	FSP 5. d); FSP 1. a); CDOS 3a Managing Resources; National FCS 2.1.1, 8.4.3	Critical path finder, Step 3	Formative; the circled step and the slip answer checked at the walk; read aloud at closure

3. Write a station card another team could follow	FSP 1. a), 1. c); CIR 3. e); CDOS 3a Managing Information; National FCS 8.4.3	Station card and the paired read, Step 4	Graded product; the flowchart and the station card are Rubric 06 criterion 3
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Homework

None required. Optional: watch anybody cook at home for ten minutes and write the order they did things in. Bring one thing they did early that you would not have thought to do early.

Connections

Inside the building: the technology or engineering teacher, if the school has one, teaches flowcharts and process diagrams with the same shapes. FACS gives that class a real process with a deadline and a consequence; the tech teacher gives FACS the standard symbols and, if willing, five minutes to look at the four posters and name the bottlenecks. Community: a restaurant kitchen manager or a caterer. FACS gives them a class that knows what a station is and what a critical path is; they give FACS a photo of a real prep list or line schedule and two minutes on what happens when one station goes down.

Sources

- NYS Middle Level CTE FACS Content Module 06, Food Systems and Production (nyctecenter.org, June 2018), lines 1. a), 1. c), and 5. d). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 02, Communication and Interpersonal Relationships, lines 3. e) and 3. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6.
- CDOS Intermediate Standard 3a, Managing Resources and Managing Information, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 2.1.1 and 8.4.3, per 02 Standards/STANDARDS SOURCES - web research.md.
- My own materials: (one page, four kitchens, five-minute blocks from setup through clean up, with a support duty column and the note that it is substitute-teacher friendly) is the exact model for the class master grid and the eight blocks. (one card per kitchen for seven students: ingredients, ten numbered steps, safety reminders, and a “what does done look like” checklist) is the exact model for the station card template.

Teacher notes

- This is the lesson that decides whether cook day works. If you have to give one lesson in this topic more time, give it this one.

- My own flowchart is for a 50-minute period. This one is eight blocks in 40 minutes, which means the cook window is roughly minutes 10 to 28 and the eat window is minutes 28 to 34. Say those numbers out loud; teams plan to fill time they do not have.
- Common mistake: teams put setup in block one and then leave blocks two and three almost empty, because they think of the recipe as a list, not a clock. The fix is the question “what is your Prep Cook doing right now,” asked at every station twice.
- Second common mistake: a station card step that is three actions in one sentence (“boil the water, add the pasta, and set the timer”). One action per numbered step. My own cards do this correctly; show step 4 of my spaghetti card as the model.
- The read-across in Step 5 finds the oven collision every year. Do not fix it for them at Step 2; let it show up on the grid so the class sees why a master flowchart exists.
- Laminate or sleeve the four station cards. They get splashed on cook day. Keep a photocopy in the folder because the sleeved one will not come back clean.